

JUMPSTART 2026 • DAY 2

When systems go wrong



gsathish.github.io/JumpStart

What to do when the work does not go to plan.

What is this problem called?

Elinor Ostrom won the 2009 Nobel Memorial Prize in Economics for her work on which problem, popularised by Garrett Hardin in 1968, in which individually rational choices destroy a shared resource?

What was this bubble called?

What name is given to the speculative bubble of the Dutch Golden Age, in which a newly fashionable flower became a luxury asset before the market collapsed in 1637?

What word names these transitions?

What word follows Mohorovičić, Gutenberg, Conrad and Lehmann to name the transition zones inside the Earth?

ACROSS BOTH ROOMS

A lot of you are worried about keeping up.

75 worries

19 workload and difficulty

15 failing or not good enough

10 falling behind

9 burnout, health, or sleep

"Insane course load."

"I'll fall behind the pace."

"Failing. Imposter syndrome. Not being good enough."

| Different words, same question: "Will I be able to keep up?"

THE OTHER HALF OF THE WALL

You also told me what helps.

THE WORRY

Workload and pace

Making friends

Keeping up

Not being good enough

WHAT HELPED

Space the work; do not cram.

Study together; ask questions.

Practise; test yourself.

Try again after a setback.

"Trying is much more important than succeeding. Don't be afraid."

TIME MANAGEMENT FOR MORTALS

4,000 weeks

Roughly an eighty-year life. There will not be time for everything.

Four Thousand Weeks

Oliver Burkeman

**Some things will have to wait.
Choose carefully.**

THE BRAESS LESSON, AGAIN

More is not always better.

Yesterday, one more road made everybody's trip longer. A study hour borrowed from sleep can make tomorrow's lecture worse.

Before you build the timetable, decide what deserves a place in it.

You cannot do everything.

You can keep showing up.

MY BORING ADVICE

- 1** Go to the lecture, tutorial, or lab.
 - 2** Try the work before you look at a solution or ask ChatGPT or another LLM.
 - 3** Come back to it on another day, in another problem.
- | Start before you are desperate, and keep coming back to the work.**

THE SIMPLEST COMMITMENT DEVICE

Attend by default.

Treat lectures as appointments, not recordings waiting in a queue.

If you miss one, decide immediately when and how you will recover it.

A COMMITMENT DEVICE

Put it in the calendar with somebody else.

Easy to ignore

"I should study physics more."

Harder to dodge

"Wednesday at 4, in the library, with Priya: two problems and one question. If I miss it, Thursday at noon."

| Decide what "done" means, and book a backup time now.

WRITE THIS DOWN

Make one plan that somebody else can check.

- 1** I will _____ at _____ with _____.
- 2** They will know I did it because _____.
- 3** If I miss it, I will _____.

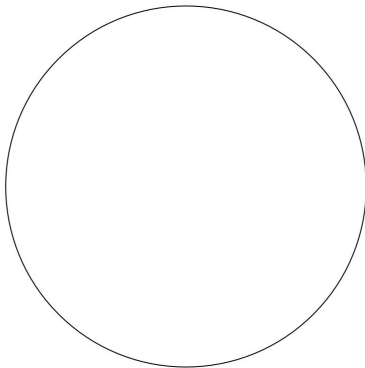
01

The research university

The people teaching you also work on unanswered questions.

A MAP OF LEARNING

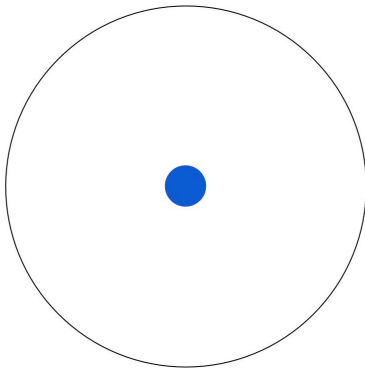
The world is much larger than any syllabus.



Adapted from Matt Might, *The illustrated guide to a Ph.D.* (CC BY-NC 2.5).

A MAP OF LEARNING

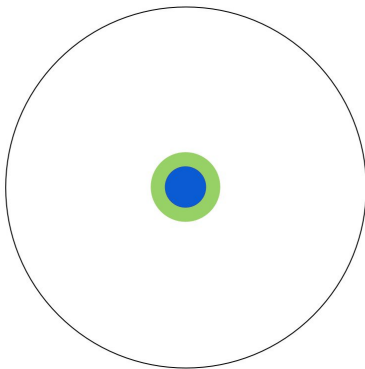
Elementary school: learn a little about many things.



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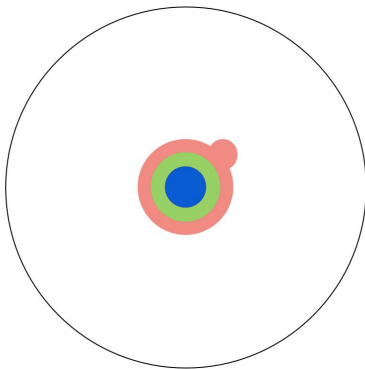
A MAP OF LEARNING

High school: add another layer of understanding.



A MAP OF LEARNING

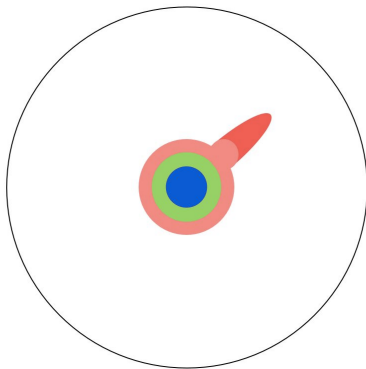
A bachelor's degree takes you wider and deeper.



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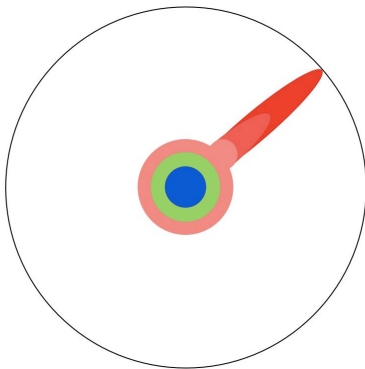
Graduate work takes you toward the edge.



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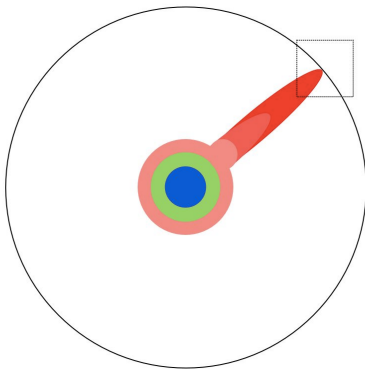
At the edge, learn the research that got there.



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A MAP OF LEARNING

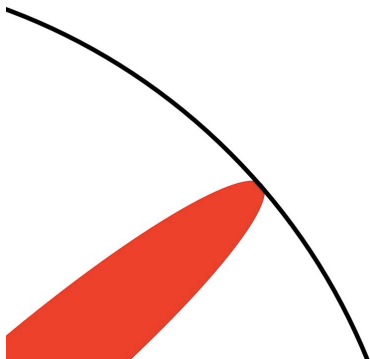
Then focus on one unanswered question.



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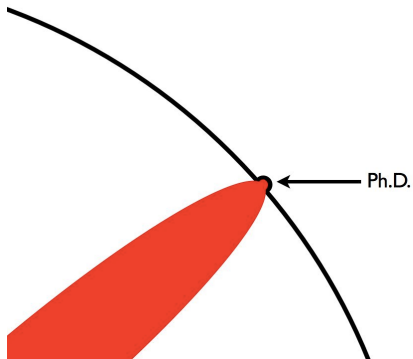
Research means pushing at that boundary.



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A MAP OF LEARNING

A Ph.D. adds one small piece to what is known.



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A MAP OF LEARNING

The bigger picture still matters.



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WHAT MAKES THIS A RESEARCH UNIVERSITY

Your instructors teach established ideas and work on questions that are still open.

In the course

The ideas, methods, and evidence you need before you can work at the boundary.

In the lab

Questions where the assumptions, tools, or answers are still unsettled.

| Lectures give you the background you need to join that work.

WHAT TO DO WITH THAT

You learn the map by showing up.

Go to the lecture because you need the argument, not only the slides. When a small gap appears, use office hours. Ask an instructor what they are trying to understand.

| First learn the map, then ask what happens at its edges.

02

Systems fail at boundaries

Nobody set out to cause harm.

ARIANE 5, FLIGHT 501 • 4 JUNE 1996

- 1** Guidance was lost 37 seconds after engine-sequence start.
The launcher broke up.

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Both inertial units ran the same software, so both failed the same way.

A BOUNDARY THE SOFTWARE DID NOT EXPECT

**The flight reached an input
the tests never covered.**

| Test the code in the conditions where it will run.

CONTENT NOTE

Therac-25, 1985–87.

A radiation therapy machine. Six known accidents caused massive overdoses, serious injuries, and deaths. This account is non-graphic.

THERAC-25 • 1985-87

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THERAC-25 • 1985-87

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- 2** Fast editing by an experienced operator could expose a race condition.
- 3** Earlier models used independent hardware interlocks. The Therac-25 relied more heavily on software for safety.

AT YOUR TABLE

- 1** List everyone who should have caught the danger.
- 2** Cross out anyone who did not know enough to catch it.

Who is left?

WHY BLAME IS HARD TO ASSIGN

The problem of many hands

Each person saw only part of the system. The patients still faced the full risk.

Moral luck

The same decision may cause harm once and go unnoticed another time. The outcome changes how we judge the engineer.

03

A convincing answer

Correct mathematics can answer the wrong question.

AUDIT THIS ANSWER

Should the city open the zero-minute link?

The total travel time is $T(y) = 4216 - 11y + y^2/4$. It is minimized at $y = 22$, giving an average trip of 60.2 minutes, below the original 62. Therefore the city should open the link: it improves travel for the population.

- What did it calculate correctly?
- What question did it fail to answer?

THE MISSING BEHAVIOUR

What it solved

A centrally assigned arrangement:
23 upper, 22 shortcut, 23 lower.

What drivers do

Leave the slower routes. The
voluntary equilibrium returns to a
68-minute trip.

The calculation assumed a traffic controller. The city has drivers.

**A polished answer can still
model the wrong thing.**

BEFORE YOU USE A LARGE LANGUAGE MODEL

Know what you need from it.

- Is an LLM allowed for this work?
- What have I tried already?
- What answer do I expect?
- How will I check what it gives me?

YOUR PART OF THE WORK

- Choose the question.
- Name the assumptions.
- Check the answer where it will be used.
- Take responsibility for what happens next.

SOME WORRIES NEED A CONVERSATION

Start with the person who can help.

- For workload or academic planning, start with Engineering Academic Services.
- For one course, ask the instructor, a TA, or the course's help room.
- For stress, sleep, or health, contact Student Health and Wellbeing.
- For connection and experience, try classmates, clubs, and design teams.

WHERE THE INTERLOCKS ARE

**For the next four years,
your interlocks are people.**

A useful question begins: "I tried X. I expected Y. I got Z."

| Name one person you will ask sooner than you want to.

**You still own
what happens next.**

WHEN YOU BUILD FOR OTHER PEOPLE

Build systems that can tell you when you are wrong.

Leave room for error and make it easy for other people to see what you missed.

| People can fix a failure once they can see it.

FOUR THOUSAND WEEKS

You cannot do everything.

Keep showing up to what matters.

Decide what matters now and make time for it.